Hydrodynamics

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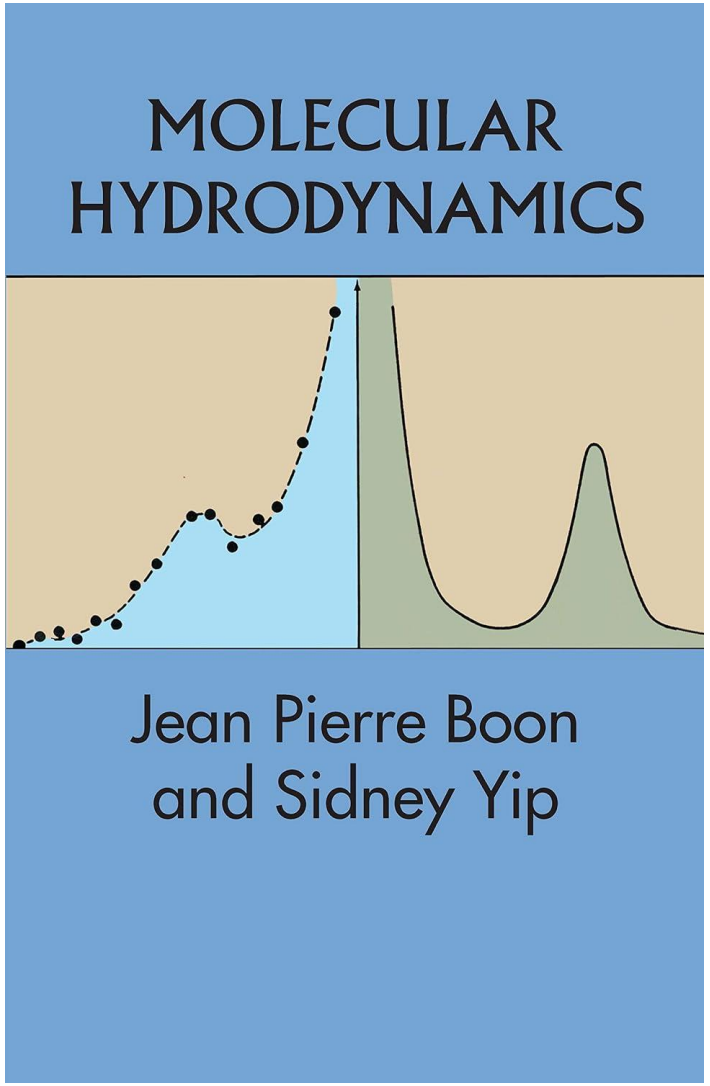
MDANSE Workshop 2026

04/11/2026



Hydrodynamics

- In this tutorial we will do something slightly different. We will how MD simulations compares hydrodynamics.
- Hydrodynamics is a macroscopy theory of a fluid which can be used to derive a scattering function equation from a correlation function of density fluctuations of the fluid.
- Its quite involved so we will only go through some of the basic ideas.



Hydrodynamics

- Consider an element of a fluid in volume V which has some property A which could be mass.

$$A(t) = \int_V \mathrm{d}^3 r \; a(\vec{r}, t)$$

Hydrodynamics

- The time derivative of this is simply the current flow of a into or out of the volume V. Assuming not sink or sources of this property.

$$\frac{\mathrm{d}A(t)}{\mathrm{d}t} = -\int_S \mathrm{d}\vec{s} \cdot \vec{J}_a(\vec{r}, t)$$

Hydrodynamics

- Gauss' theorem.

$$\int_V \mathrm{d}^3 r \left[\frac{\partial a(\vec{r}, t)}{\partial t} + \vec{\nabla} \cdot \vec{J}_a(\vec{r}, t) \right] = 0$$

Hydrodynamics

- Since the volume is arbitrary.

$$\frac{\partial a(\vec{r},t)}{\partial t} + \vec{\nabla} \cdot \vec{J}_a(\vec{r},t) = 0$$

Hydrodynamics

- We can apply the conservation equation to the properties of mass, momentum, and energy.

Mass $$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot [\rho \vec{v}] = 0$$

Momenta $$\frac{\partial \rho \vec{v}}{\partial t} + \vec{\nabla} \cdot [\rho \vec{v} \otimes \vec{v} - \overleftrightarrow{\sigma}] = 0$$

Energy $$\frac{\partial e}{\partial t} + \vec{\nabla} \cdot \left[e\vec{v} + \vec{Q} - \vec{v} \cdot \overleftrightarrow{\sigma} \right] = 0$$

Hydrodynamics

- We can apply the conservation equation to the properties of mass, momentum, and energy.
- Using the mass continuity equation, we can obtain Cauchy momentum equation.

Mass $$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot [\rho \vec{v}] = 0$$

Momenta $$\rho \left[\frac{\partial \vec{v}}{\partial t} + \left(\vec{v} \cdot \vec{\nabla} \right) \vec{v} \right] - \vec{\nabla} \cdot \overleftrightarrow{\sigma} = 0$$

Energy $$\frac{\partial e}{\partial t} + \vec{\nabla} \cdot \left[e\vec{v} + \vec{Q} - \vec{v} \cdot \overleftrightarrow{\sigma} \right] = 0$$

Hydrodynamics

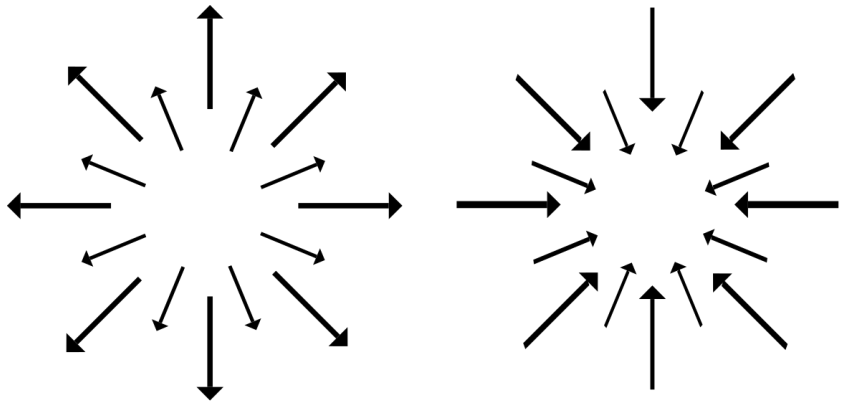
- Let's look more closely to see what the equations are saying.

Mass

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot [\rho \vec{v}] = 0$$

mass flux

Divergence



$\partial/\partial x(\mathbf{V}_x) > 0$
$\partial/\partial y(\mathbf{V}_y) > 0$
$\nabla \bullet (\mathbf{V}) > 0$

$\partial/\partial x(\mathbf{V}_x) < 0$
$\partial/\partial y(\mathbf{V}_y) < 0$
$\nabla \bullet (\mathbf{V}) < 0$

$$\nabla \cdot \mathbf{V}(x,y) = \frac{\partial V_x(x,y)}{\partial x} + \frac{\partial V_y(x,y)}{\partial y}$$

Hydrodynamics

- Let's look more closely to see what the equations are saying.

Momenta

$$\rho\left[\frac{\partial\vec{v}}{\partial t}+\left(\vec{v}\cdot\vec{\nabla}\right)\vec{v}\right]-\vec{\nabla}\cdot\overleftrightarrow{\sigma}=0$$

material derivative

total derivative with respect to time

$$\frac{\mathrm{D}\vec{v}}{\mathrm{D}t}=\frac{\mathrm{d}\vec{v}}{\mathrm{d}t}=\frac{\partial\vec{v}}{\partial t}+\frac{\partial\vec{v}}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t}+\frac{\partial\vec{v}}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}+\frac{\partial\vec{v}}{\partial z}\frac{\mathrm{d}z}{\mathrm{d}t}$$

$$=\frac{\partial\vec{v}}{\partial t}+\left(\vec{v}\cdot\nabla\right)\vec{v}$$

Hydrodynamics

- Let's look more closely to see what the equations are saying.

Momenta

$$\rho\left[\frac{\partial\vec{v}}{\partial t}+\left(\vec{v}\cdot\vec{\nabla}\right)\vec{v}\right]-\vec{\nabla}\cdot\overleftrightarrow{\sigma}=0$$

Cauchy stress tensor

$$\begin{bmatrix}\sigma_{xx} & \sigma_{xy} & \sigma_{xz}\\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz}\\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz}\end{bmatrix}$$

shear components

normal components

total force on volume V

$$\int_V \mathrm{d}^3r\ \vec{\nabla}\cdot\overleftrightarrow{\sigma}=\int_S \mathrm{d}\vec{s}\cdot\overleftrightarrow{\sigma}$$

Hydrodynamics

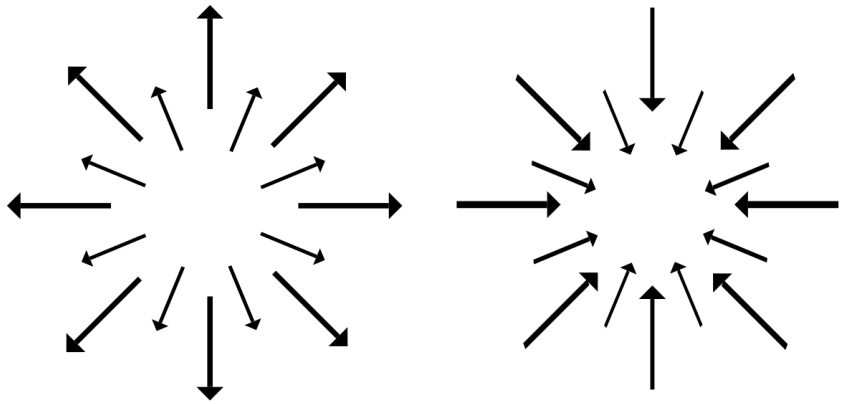
- Let's look more closely to see what the equations are saying.

Energy

$$\frac{\partial e}{\partial t} + \vec{\nabla} \cdot \left[e\vec{v} + \vec{Q} - \vec{v} \cdot \overleftrightarrow{\sigma} \right] = 0$$

energy flux
(local kinetic and
internal energy)

Divergence



$\partial/\partial x(\mathbf{V}_x) > 0$
$\partial/\partial y(\mathbf{V}_y) > 0$
$\nabla \bullet (\mathbf{V}) > 0$

$\partial/\partial x(\mathbf{V}_x) < 0$
$\partial/\partial y(\mathbf{V}_y) < 0$
$\nabla \bullet (\mathbf{V}) < 0$

$$\nabla \cdot \mathbf{V}(x,y) = \frac{\partial V_x(x,y)}{\partial x} + \frac{\partial V_y(x,y)}{\partial y}$$

Hydrodynamics

- Let's look more closely to see what the equations are saying.

Energy

$$\frac{\partial e}{\partial t} + \vec{\nabla} \cdot \left[e\vec{v} + \vec{Q} - \vec{v} \cdot \overleftrightarrow{\sigma} \right] = 0$$

heat flux

Fourier's law

$$\vec{Q} = -\lambda \vec{\nabla} T$$

flux of heat energy density

thermal conductivity

temperature

Hydrodynamics

- Let's look more closely to see what the equations are saying.

Energy

$$\frac{\partial e}{\partial t} + \vec{\nabla} \cdot \left[e\vec{v} + \vec{Q} - \vec{v} \cdot \overleftrightarrow{\sigma} \right] = 0$$

work done on volume V per unit time

$$\int_V \mathrm{d}^3 r \; \vec{\nabla} \cdot [\vec{v} \cdot \overleftrightarrow{\sigma}] = \int_S \mathrm{d}\vec{s} \cdot [\vec{v} \cdot \overleftrightarrow{\sigma}]$$

Hydrodynamics

- Cauchy stress tensor for a compressible Newtonian fluid.

$$\overleftrightarrow{\sigma} = -p\overleftrightarrow{I} + \eta_s\left[\vec{\nabla}\otimes\vec{v} + (\vec{\nabla}\otimes\vec{v})^{\mathrm{T}} - \frac{2}{3}(\vec{\nabla}\cdot\vec{v})\overleftrightarrow{I}\right] + \eta_v(\vec{\nabla}\cdot\vec{v})\overleftrightarrow{I}$$

Cauchy stress tensor

pressure density

shear viscosity

bulk viscosity

Hydrodynamics

- Cauchy stress tensor for a compressible Newtonian fluid.

Navier-Stokes Equation

$$\overleftrightarrow{\sigma} = -p\overleftrightarrow{I} + \eta_s\left[\vec{\nabla}\otimes\vec{v} + (\vec{\nabla}\otimes\vec{v})^{\mathrm{T}} - \frac{2}{3}(\vec{\nabla}\cdot\vec{v})\overleftrightarrow{I}\right] + \eta_v(\vec{\nabla}\cdot\vec{v})\overleftrightarrow{I}$$

$$\rho\left[\frac{\partial\vec{v}}{\partial t} + \left(\vec{v}\cdot\vec{\nabla}\right)\vec{v}\right] - \vec{\nabla}\cdot\overleftrightarrow{\sigma} = 0$$

Hydrodynamics

- Cauchy stress tensor for an incompressible Newtonian fluid.

Navier-Stokes Equation

$$\overleftrightarrow{\sigma} = -p\overleftrightarrow{I} + \eta_s\left[\vec{\nabla}\otimes\vec{v} + (\vec{\nabla}\otimes\vec{v})^{\mathrm{T}}\right] \qquad \vec{\nabla}\cdot\vec{v} = 0$$

$$\rho\left[\frac{\partial\vec{v}}{\partial t} + \left(\vec{v}\cdot\vec{\nabla}\right)\vec{v}\right] - \vec{\nabla}\cdot\overleftrightarrow{\sigma} = 0$$

Hydrodynamics

- Cauchy stress tensor for an incompressible Newtonian fluid.

Navier-Stokes Equation

$$\rho\left[\frac{\partial\vec{v}}{\partial t}+\left(\vec{v}\cdot\vec{\nabla}\right)\vec{v}\right]=-\vec{\nabla}p+\eta_s\nabla^2\vec{v}$$

$$\vec{\nabla}\cdot\vec{v}=0$$

Hydrodynamics

- Cauchy stress tensor for an incompressible Newtonian fluid.

Navier-Stokes Equation

$$\frac{\partial \vec{v}}{\partial t} + \left(\vec{v} \cdot \vec{\nabla}\right) \vec{v} + \vec{\nabla} p - \eta_s \nabla^2 \vec{v} = \vec{f}$$

$$\vec{\nabla} \cdot \vec{v} = 0$$

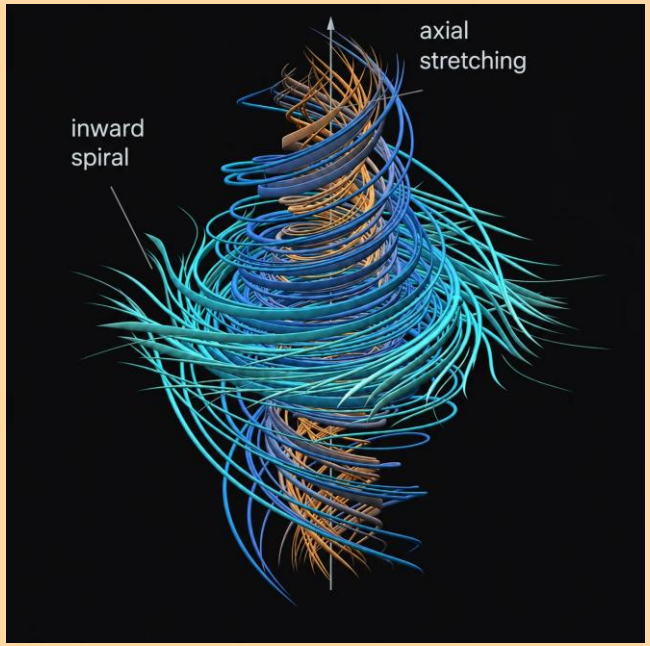
FINITE TIME BLOWUP FOR NAVIER–STOKES

OPENAI

ABSTRACT. For every positive viscosity, we construct a solution of the three-dimensional incompressible Navier–Stokes equations that starts from rest and develops unbounded velocity in finite time while maintaining uniformly bounded kinetic energy.

Theorem 1.1. *For every $\nu > 0$ there exist a force $f \in C_c^\infty(\mathbb{R}^3 \times (0,\infty); \mathbb{R}^3)$, a compact set $K \subset \mathbb{R}^3$, and smooth velocity and pressure fields u, p on $\mathbb{R}^3 \times [0,1)$ satisfying*

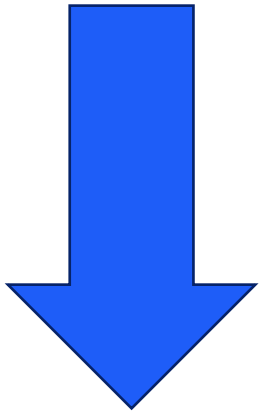
$$\partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p = f, \qquad \nabla \cdot u = 0, \qquad u(\cdot, 0) = 0, \tag{1.1}$$



Hydrodynamics

$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot [\rho \vec{v}] = 0 \qquad \rho \left[\frac{\partial \vec{v}}{\partial t} + \left(\vec{v} \cdot \vec{\nabla} \right) \vec{v} \right] - \vec{\nabla} \cdot \overleftrightarrow{\sigma} = 0 \qquad \frac{\partial e}{\partial t} + \vec{\nabla} \cdot \left[e\vec{v} + \vec{Q} - \vec{v} \cdot \overleftrightarrow{\sigma} \right] = 0$$

$$\rho = \rho_0 + \rho_1$$
$$\vec{v} = \vec{v}_0 + \vec{v}_1 = \vec{v}_1$$
$$e = e_0 + e_1$$
$$T = T_0 + T_1$$



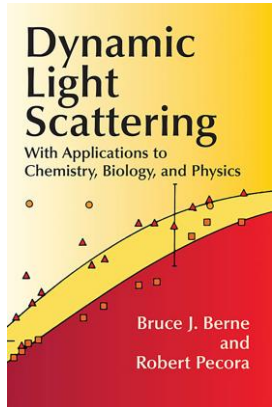
Linearised hydrodynamics

$$\frac{\partial \rho_1}{\partial t} + \rho_0 \vec{\nabla} \cdot \vec{v} = 0 \qquad \rho_0 \frac{\partial \vec{v}}{\partial t} = -\vec{\nabla} p_1 + \eta_s \nabla^2 \vec{v} + \left(\eta_v + \frac{1}{3} \eta_s \right) \vec{\nabla} (\vec{\nabla} \cdot \vec{v}) \qquad \frac{\partial e_1}{\partial t} + (e_0 + p_0) \vec{\nabla} \cdot \vec{v} = \lambda \nabla^2 T_1$$

Hydrodynamics

$$\frac{\partial \rho_1}{\partial t} + \rho_0 \vec{\nabla} \cdot \vec{v} = 0 \qquad \rho_0 \frac{\partial \vec{v}}{\partial t} = -\vec{\nabla} p_1 + \eta_s \nabla^2 \vec{v} + \left(\eta_v + \frac{1}{3}\eta_s\right) \vec{\nabla}(\vec{\nabla} \cdot \vec{v}) \qquad \frac{\partial e_1}{\partial t} + (e_0 + p_0)\vec{\nabla} \cdot \vec{v} = \lambda \nabla^2 T_1$$

density fluctuation correlation function



Lots of maths
pages 237 to 243

$$S(q,\omega) = \frac{1}{\pi} V \rho k_{\mathrm{B}} T \chi_T \left\{ \left(\frac{\gamma - 1}{\gamma}\right) \frac{D_T q^2}{\omega^2 + (D_T q^2)^2} + \frac{1}{\gamma} \left[\frac{\Gamma q^2}{(\omega + c_{\mathrm{s}} q)^2 + (\Gamma q^2)^2} + \frac{\Gamma q^2}{(\omega - c_{\mathrm{s}} q)^2 + (\Gamma q^2)^2} \right] + \cdots \right\}$$

Hydrodynamics

Central Rayleigh peak

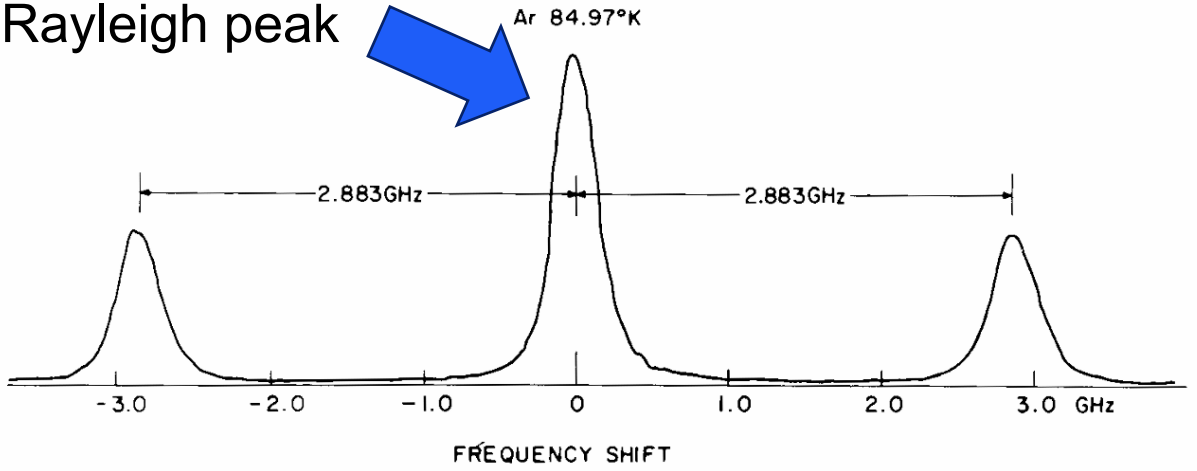
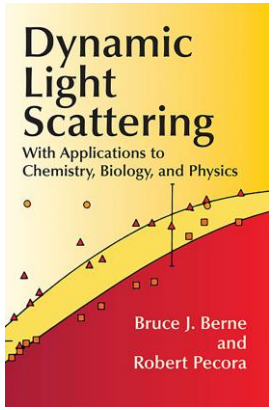


FIG. 10.4.1. Brillouin spectrum of liquid argon ($T = 84.97°\text{K}$, $\theta = 90° 14'$, laser wavelength = 5145 Å). (From Fleury and Boon, 1969.)



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$$S(q,\omega) = \frac{1}{\pi} V \rho k_{\mathrm{B}} T \chi_T \left\{ \left(\frac{\gamma - 1}{\gamma} \right) \frac{D_T q^2}{\omega^2 + (D_T q^2)^2} + \frac{1}{\gamma} \left[\frac{\Gamma q^2}{(\omega + c_{\mathrm{s}} q)^2 + (\Gamma q^2)^2} + \frac{\Gamma q^2}{(\omega - c_{\mathrm{s}} q)^2 + (\Gamma q^2)^2} \right] + \cdots \right\}$$

Hydrodynamics

Brillouin doublet

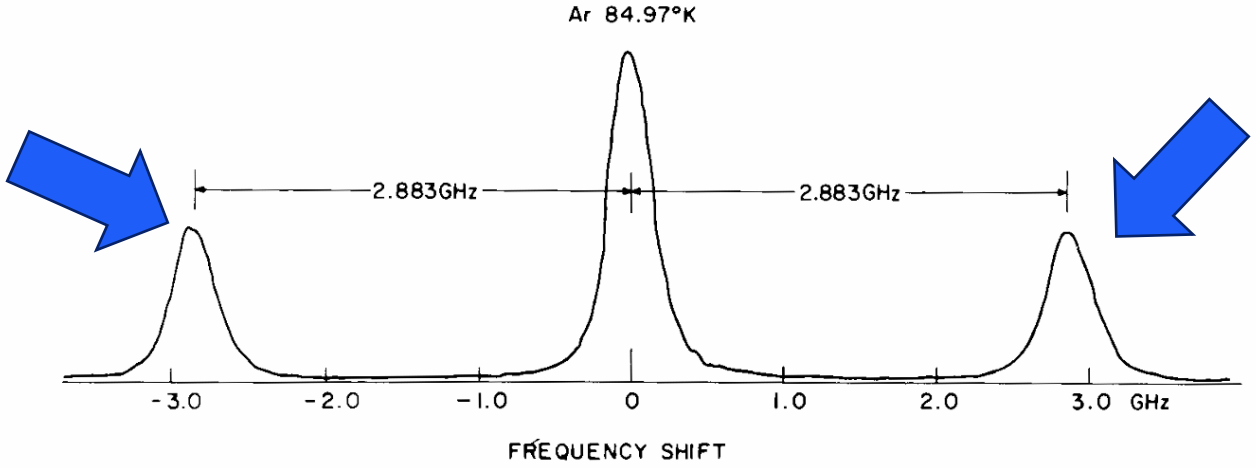
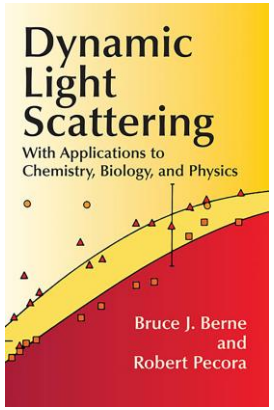


FIG. 10.4.1. Brillouin spectrum of liquid argon ($T = 84.97°\text{K}$, $\theta = 90° 14'$, laser wavelength = 5145 Å). (From Fleury and Boon, 1969.)



Page 244

$$S(q,\omega) = \frac{1}{\pi} V \rho k_{\mathrm{B}} T \chi_T \left\{ \left(\frac{\gamma - 1}{\gamma} \right) \frac{D_T q^2}{\omega^2 + (D_T q^2)^2} + \frac{1}{\gamma} \left[\frac{\Gamma q^2}{(\omega + c_{\mathrm{s}} q)^2 + (\Gamma q^2)^2} + \frac{\Gamma q^2}{(\omega - c_{\mathrm{s}} q)^2 + (\Gamma q^2)^2} \right] + \cdots \right\}$$

Hydrodynamics

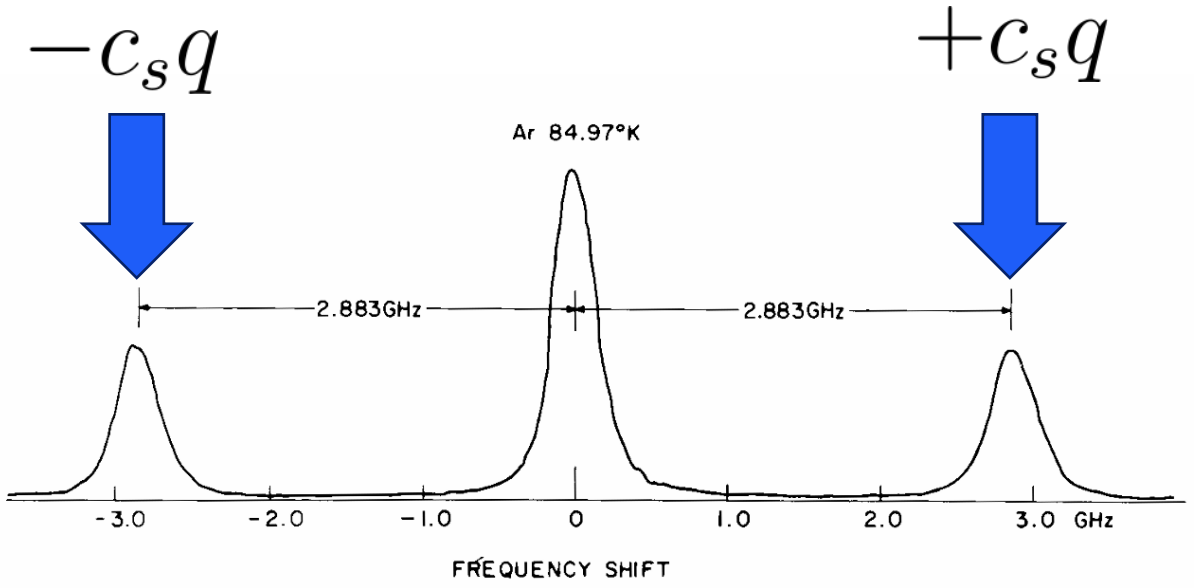
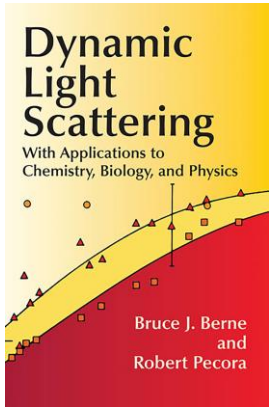


FIG. 10.4.1. Brillouin spectrum of liquid argon ($T = 84.97°\text{K}$, $\theta = 90° 14'$, laser wavelength = 5145 Å). (From Fleury and Boon, 1969.)



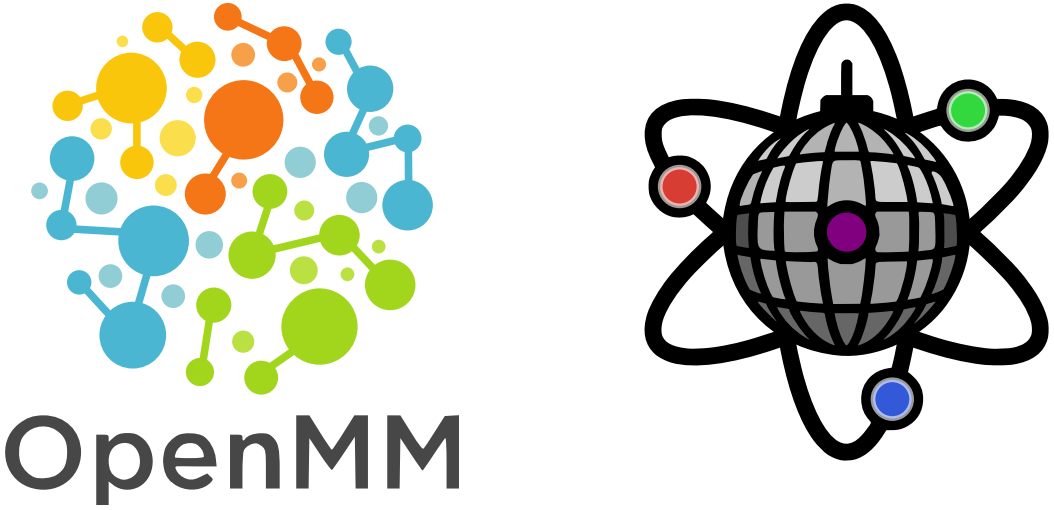
Page 244

$$S(q,\omega) = \frac{1}{\pi} V \rho k_{\mathrm{B}} T \chi_T \left\{ \left(\frac{\gamma - 1}{\gamma} \right) \frac{D_T q^2}{\omega^2 + (D_T q^2)^2} + \frac{1}{\gamma} \left[\frac{\Gamma q^2}{(\omega + c_{\mathrm{s}} q)^2 + (\Gamma q^2)^2} + \frac{\Gamma q^2}{(\omega - c_{\mathrm{s}} q)^2 + (\Gamma q^2)^2} \right] + \cdots \right\}$$

sound speed

Argon

- There is a script named `argon.py` which you can use to generate argon trajectories for some of the next tutorials.
- It's quite similar to the OpenMM water script that you ran earlier.
- Again, this script is incomplete, and you will need to finish it off to get it working.
- This script is very similar to the one you used to create the water trajectory. So, we won't go through every part of the script.



Argon

- As with the water OpenMM script, if you see … this is where you will need to fill in to complete the script.
- We will do two run one with argon-36 and another with natural-abundance argon. Makes sure to set a good file name and the correct mass.

```
import itertools as it
import sys

import openmm as mm
from openmm import app, unit


OUTPUT_DCD_NAME = ...
OUTPUT_PDB_NAME = ...

TEMPERATURE = 85.0 * unit.kelvin
FRICTION = 1.0 / unit.picosecond
TIMESTEP = 2.0 * unit.femtosecond

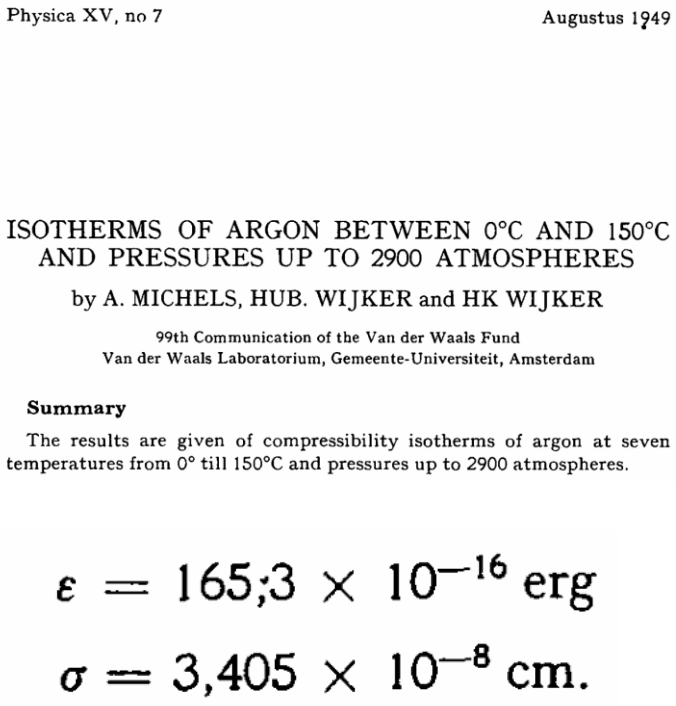
NVT_STEPS = 12000
NVE_EQUIL_STEPS = 12000
NVE_PROD_STEPS = 60060
REPORT_INTERVAL = 60

MASS = ... * u

N_CELLS = 4
CELL_LEN = 0.57311  # nm
BOX_LEN = N_CELLS * CELL_LEN  # nm
LJ_SIGMA = ...
LJ_EPS = ...
```

Argon

- For the Lennard-Jones parameters we will use the same values mentioned in



Physica XV, no 7 Augustus 1949

ISOTHERMS OF ARGON BETWEEN 0°C AND 150°C AND PRESSURES UP TO 2900 ATMOSPHERES

by A. MICHELS, HUB. WIJKER and HK WIJKER

99th Communication of the Van der Waals Fund
Van der Waals Laboratorium, Gemeente-Universiteit, Amsterdam

Summary

The results are given of compressibility isotherms of argon at seven temperatures from 0° till 150°C and pressures up to 2900 atmospheres.

$$\varepsilon = 165{;}3 \times 10^{-16} \text{ erg}$$

$$\sigma = 3{,}405 \times 10^{-8} \text{ cm.}$$

```python
import itertools as it
import sys

import openmm as mm
from openmm import app, unit


OUTPUT_DCD_NAME = ...
OUTPUT_PDB_NAME = ...

TEMPERATURE = 85.0 * unit.kelvin
FRICTION = 1.0 / unit.picosecond
TIMESTEP = 2.0 * unit.femtosecond

NVT_STEPS = 12000
NVE_EQUIL_STEPS = 12000
NVE_PROD_STEPS = 60060
REPORT_INTERVAL = 60

MASS = ... * unit.amu

N_CELLS = 4
CELL_LEN = 0.57311  # nm
BOX_LEN = N_CELLS * CELL_LEN  # nm
LJ_SIGMA = ...
LJ_EPS = ...
```

Argon

- This time we are not using packmol, instead we use OpenMM, to create a box of argon atoms in an FCC lattice.

```
# create topology and set positions
fcc_basis = [(0.0, 0.0, 0.0), (0.5, 0.5, 0.0), (0.5, 0.0, 0.5), (0.0, 0.5, 0.5)]
topology = app.Topology()
chain = topology.addChain()
argon = app.Element.getBySymbol("Ar")

positions = []
for (i, j, k), (bx, by, bz) in it.product(
    it.product(range(N_CELLS), repeat=3), fcc_basis
):
    residue = topology.addResidue("AR", chain)
    topology.addAtom("Ar", argon, residue)
    positions.append(mm.Vec3(i + bx, j + by, k + bz) * CELL_LEN)
positions = unit.Quantity(positions, unit.nanometer)

topology.setPeriodicBoxVectors(
    (
        mm.Vec3(BOX_LEN, 0, 0),
        mm.Vec3(0, BOX_LEN, 0),
        mm.Vec3(0, 0, BOX_LEN)
    ) * unit.nanometer
)
```

Argon

- Next, we create the OpenMM system and a nonbonded force object. See https://docs.openmm.org/latest/userguide/theory/02_standard_forces.html#nonbondedforce for more details.
- Nonbonded force it's a combination of a Lennard-Jones and coulomb interaction. The argon atoms are neutral so we can ignore the coulomb interaction.

$$E = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^{6} \right]$$

```
52  # create openmm system, and create and add forces
53  system = mm.System()
54  system.setDefaultPeriodicBoxVectors(*topology.getPeriodicBoxVectors())
55
56  force = mm.NonbondedForce()
57  force.setNonbondedMethod(mm.NonbondedForce.CutoffPeriodic)
58  force.setCutoffDistance(1.0 * unit.nanometer)
59  force.setUseSwitchingFunction(True)
60  force.setSwitchingDistance(0.9 * unit.nanometer)
61  force.setUseDispersionCorrection(True)
62
63  for _ in range(topology.getNumAtoms()):
64      system.addParticle(MASS)
65      force.addParticle(...)
66
67  system.addForce(force)
```

Argon

- Here we use cutoff periodic.

Periodic boundary conditions are used, so that each particle interacts only with the nearest periodic copy of each other particle. Interactions beyond the cutoff distance are ignored. Coulomb interactions closer than the cutoff distance are modified using the reaction field method.

https://docs.openmm.org/development/api-c++/generated/NonbondedForce.html

```
52  # create openmm system, and create and add forces
53  system = mm.System()
54  system.setDefaultPeriodicBoxVectors(*topology.getPeriodic[illegible]ctors())
55
56  force = mm.NonbondedForce()
57  force.setNonbondedMethod(mm.NonbondedForce.CutoffPeriodic)
58  force.setCutoffDistance(1.0 * unit.nanometer)
59  force.setUseSwitchingFunction(True)
60  force.setSwitchingDistance(0.9 * unit.nanometer)
61  force.setUseDispersionCorrection(True)
62
63  for _ in range(topology.getNumAtoms()):
64      system.addParticle(MASS)
65      force.addParticle(...)
66
67  system.addForce(force)
```

Argon

- Here we use cutoff periodic.

> *Periodic boundary conditions are used, so that each particle interacts only with the nearest periodic copy of each other particle. Interactions beyond the cutoff distance are ignored. Coulomb interactions closer than the cutoff distance are modified using the reaction field method.*

https://docs.openmm.org/development/api-c++/generated/NonbondedForce.html

- And the switching function which smoothly reduces the interaction to zero before the cutoff.

```
52  # create openmm system, and create and add forces
53  system = mm.System()
54  system.setDefaultPeriodicBoxVectors(*topology.getPeriodicBoxVectors())
55
56  force = mm.NonbondedForce()
57  force.setNonbondedMethod(mm.NonbondedForce.CutoffPeriodic)
58  force.setCutoffDistance(1.0 * unit.nan
59  force.setUseSwitchingFunction(True)
60  force.setSwitchingDistance(0.9 * unit.nanometer)
61  force.setUseDispersionCorrection(True)
62
63  for _ in range(topology.getNumAtoms()):
64      system.addParticle(MASS)
65      force.addParticle(...)
66
67  system.addForce(force)
```

Argon

- Here we use cutoff periodic.

> *Periodic boundary conditions are used, so that each particle interacts only with the nearest periodic copy of each other particle. Interactions beyond the cutoff distance are ignored. Coulomb interactions closer than the cutoff distance are modified using the reaction field method.*
>
> https://docs.openmm.org/development/api-c++/generated/NonbondedForce.html

- And the switching function which smoothly reduces the interaction to zero before the cutoff.
- Add a long-range correction to approximate the interaction beyond the cutoff.

```
52  # create openmm system, and create and add forces
53  system = mm.System()
54  system.setDefaultPeriodicBoxVectors(*topology.getPeriodicBoxVectors())
55
56  force = mm.NonbondedForce()
57  force.setNonbondedMethod(mm.NonbondedForce.CutoffPeriodic)
58  force.setCutoffDistance(1.0 * unit.nanometer)
59  force.setUseSwitchingFunction(True)
60  force.setSwitchingDistance(0.9 * unit.nan
61  force.setUseDispersionCorrection(True)
62
63  for _ in range(topology.getNumAtoms()):
64      system.addParticle(MASS)
65      force.addParticle(...)
66
67  system.addForce(force)
```

$$E_{\text{cor}} = \frac{8\pi N^2}{V}\left(\frac{\langle \epsilon_{ij}\sigma_{ij}^{12}\rangle}{9{r_c}^9} - \frac{\langle \epsilon_{ij}\sigma_{ij}^{6}\rangle}{3{r_c}^3}\right)$$

Argon

- Finish this part of the code. Check https://docs.openmm.org/latest/api-python/generated/openmm.openmm.NonbondedForce.html.

```
52  # create openmm system, and create and add forces
53  system = mm.System()
54  system.setDefaultPeriodicBoxVectors(*topology.getPeriodicBoxVectors())
55
56  force = mm.NonbondedForce()
57  force.setNonbondedMethod(mm.NonbondedForce.CutoffPeriodic)
58  force.setCutoffDistance(1.0 * unit.nanometer)
59  force.setUseSwitchingFunction(True)
60  force.setSwitchingDistance(0.9 * unit.nanometer)
61  force.setUseDispersionCorrection(True)
62
63  for _ in range(topology.getNumAtoms()):
64      system.addParticle(MASS)
65      force.addParticle(...)
66
67  system.addForce(force)
```



Argon

- Finish this part of the code. Check https://docs.openmm.org/latest/api-python/generated/openmm.openmm.NonbondedForce.html.
- Here we add our nonbonded force to the system.

```
52  # create openmm system, and create and add forces
53  system = mm.System()
54  system.setDefaultPeriodicBoxVectors(*topology.getPeriodicBoxVectors())
55
56  force = mm.NonbondedForce()
57  force.setNonbondedMethod(mm.NonbondedForce.CutoffPeriodic)
58  force.setCutoffDistance(1.0 * unit.nanometer)
59  force.setUseSwitchingFunction(True)
60  force.setSwitchingDistance(0.9 * unit.nanometer)
61  force.setUseDispersionCorrection(True)
62
63  for _ in range(topology.getNumAtoms()):
64      system.addParticle(MASS)
65      force.addParticle(...)
66
67  system.addForce(force)
```



Argon

- The rest of the code is very similar to the water script, but we skip the NPT equilibration stage.
- Once you are done run the script to generate trajectory files for argon-36 and natural-abundance argon.
- Convert both the argon trajectory with the DCD converter, as done in the water tutorials. Make sure you set the correct time step.

```python
# create the langevin and verlet integrators
nvt_integrator = mm.LangevinMiddleIntegrator(TEMPERATURE, FRICTION, TIMESTEP)
nve_integrator = mm.VerletIntegrator(TIMESTEP)
integrator = mm.CompoundIntegrator()
integrator.addIntegrator(nvt_integrator)
integrator.addIntegrator(nve_integrator)
integrator.setCurrentIntegrator(0)

# create the openmm simulation and set the positions and velocities
simulation = app.Simulation(topology, system, integrator)
simulation.context.setPositions(positions)
simulation.context.setVelocitiesToTemperature(TEMPERATURE)

# create some reporters which outputs some details of the simulation
simulation.reporters.append(app.StateDataReporter(
    sys.stdout, REPORT_INTERVAL, step=True, temperature=True,
    density=True, speed=True))

# minimisation
print(f"Running OpenMM under the platform: {simulation.context.getPlatform().getName()}")
simulation.minimizeEnergy()

# NVT equilibration
print(f"Running NVT equilibration for: {NVT_STEPS} steps")
simulation.step(NVT_STEPS)
simulation.reporters.clear()

# NVE equilibration
print(f"Running NVE equilibration for: {NVE_EQUIL_STEPS} steps")
integrator.setCurrentIntegrator(1)
simulation.step(NVE_EQUIL_STEPS)

# NVE production
print(f"Running NVE production for: {NVE_PROD_STEPS} steps")
# clear and create new reporters for the production stage
simulation.reporters.clear()
simulation.reporters.append(app.DCDReporter(OUTPUT_DCD_NAME, REPORT_INTERVAL))
simulation.reporters.append(app.StateDataReporter(
    sys.stdout, REPORT_INTERVAL, step=True, temperature=True,
    totalEnergy=True, speed=True, totalSteps=NVE_PROD_STEPS,
    progress=True))
# reset the time back to zero and run the production stage
simulation.context.setStepCount(0)
simulation.context.setTime(0 * unit.picosecond)
simulation.step(NVE_PROD_STEPS)

# Save the final structure. This is the topology file needed to the trajectory
# conversion with the DCD converter.
state = simulation.context.getState(getPositions=True, enforcePeriodicBox=True)
with open(OUTPUT_PDB_NAME, "w") as f:
    app.PDBFile.writeFile(simulation.topology, state.getPositions(), f)
```

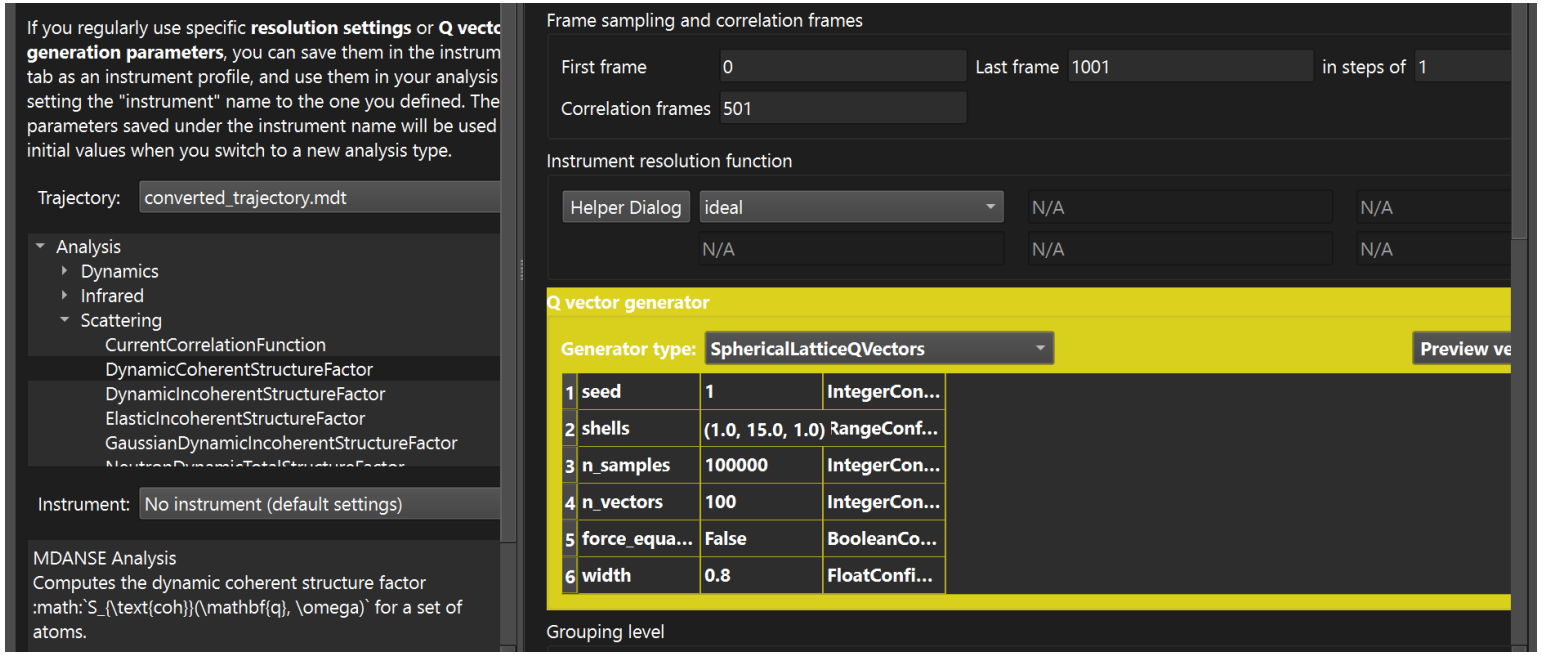
Hydrodynamics

- Lets try to see if we get the anything at all similar to what hydrodynamics suggests for the scattering function from molecular dynamics.
- Load up the natural-abundance argon trajectory and select the dynamic coherent structure factor job.

$$S(q,\omega) = \frac{1}{\pi} V \rho k_{\mathrm{B}} T \chi_T \left\{ \left(\frac{\gamma - 1}{\gamma} \right) \frac{D_T q^2}{\omega^2 + (D_T q^2)^2} + \frac{1}{\gamma} \left[\frac{\Gamma q^2}{(\omega + c_{\mathrm{s}} q)^2 + (\Gamma q^2)^2} + \frac{\Gamma q^2}{(\omega - c_{\mathrm{s}} q)^2 + (\Gamma q^2)^2} \right] + \cdots \right\}$$

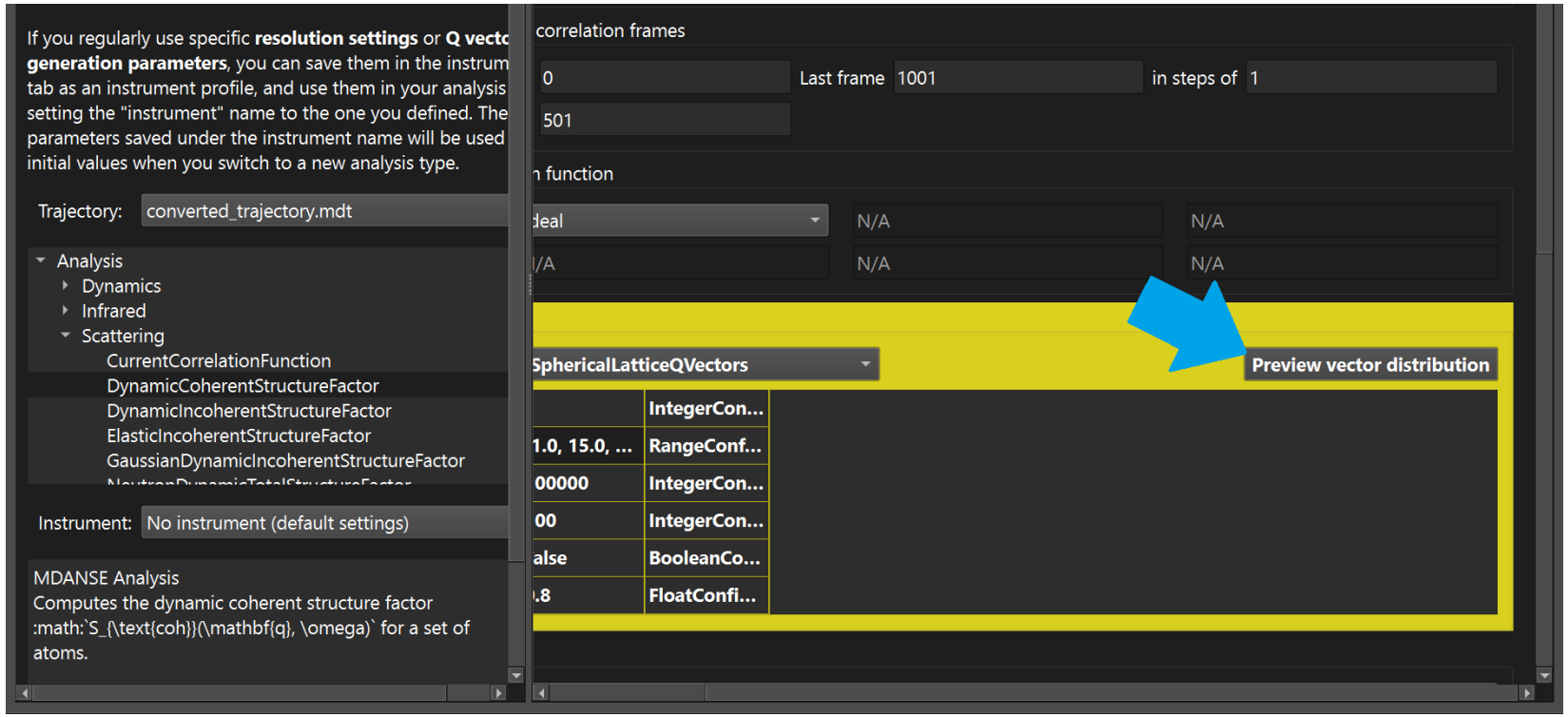
Hydrodynamics

- Set the q-vector generator with the following setting. Notice that the widget has highlighted yellow.



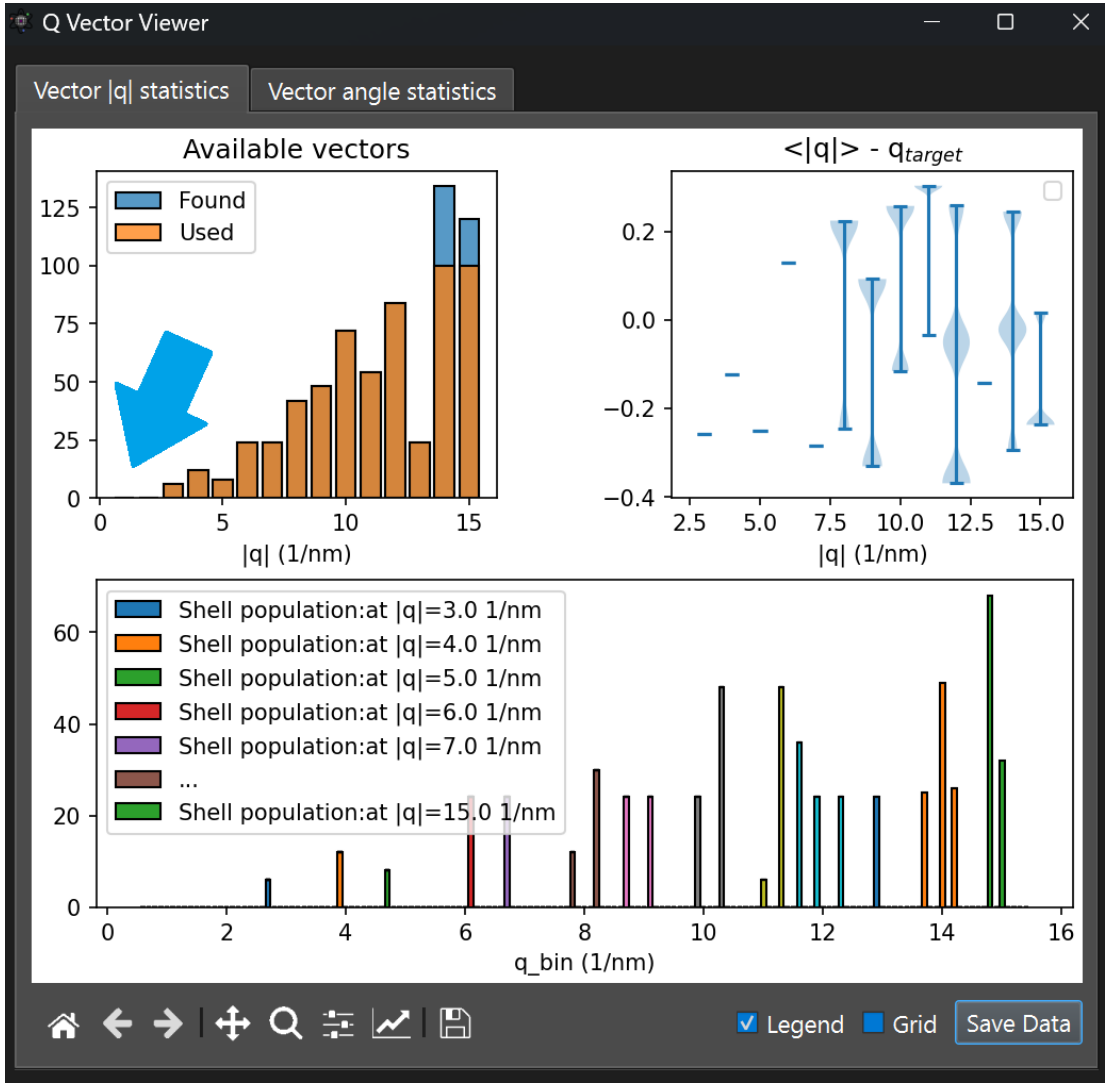
Hydrodynamics

- Open the preview vector distribution so see what's going on.



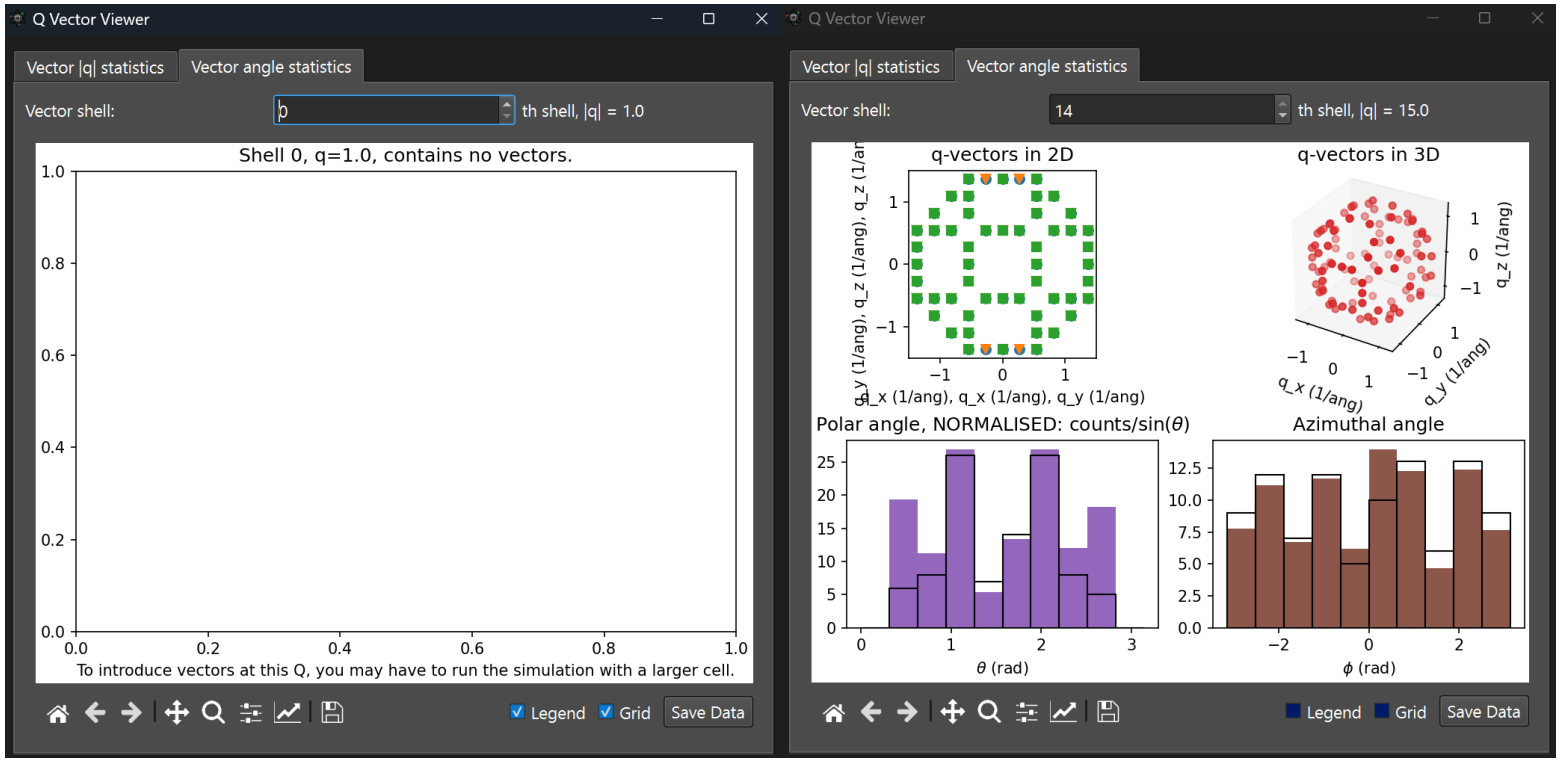
Hydrodynamics

- You should see that the smaller q-vectors shell were not filled. The unit cell of our system was too small.
- There are no reciprocal lattice points in that range.



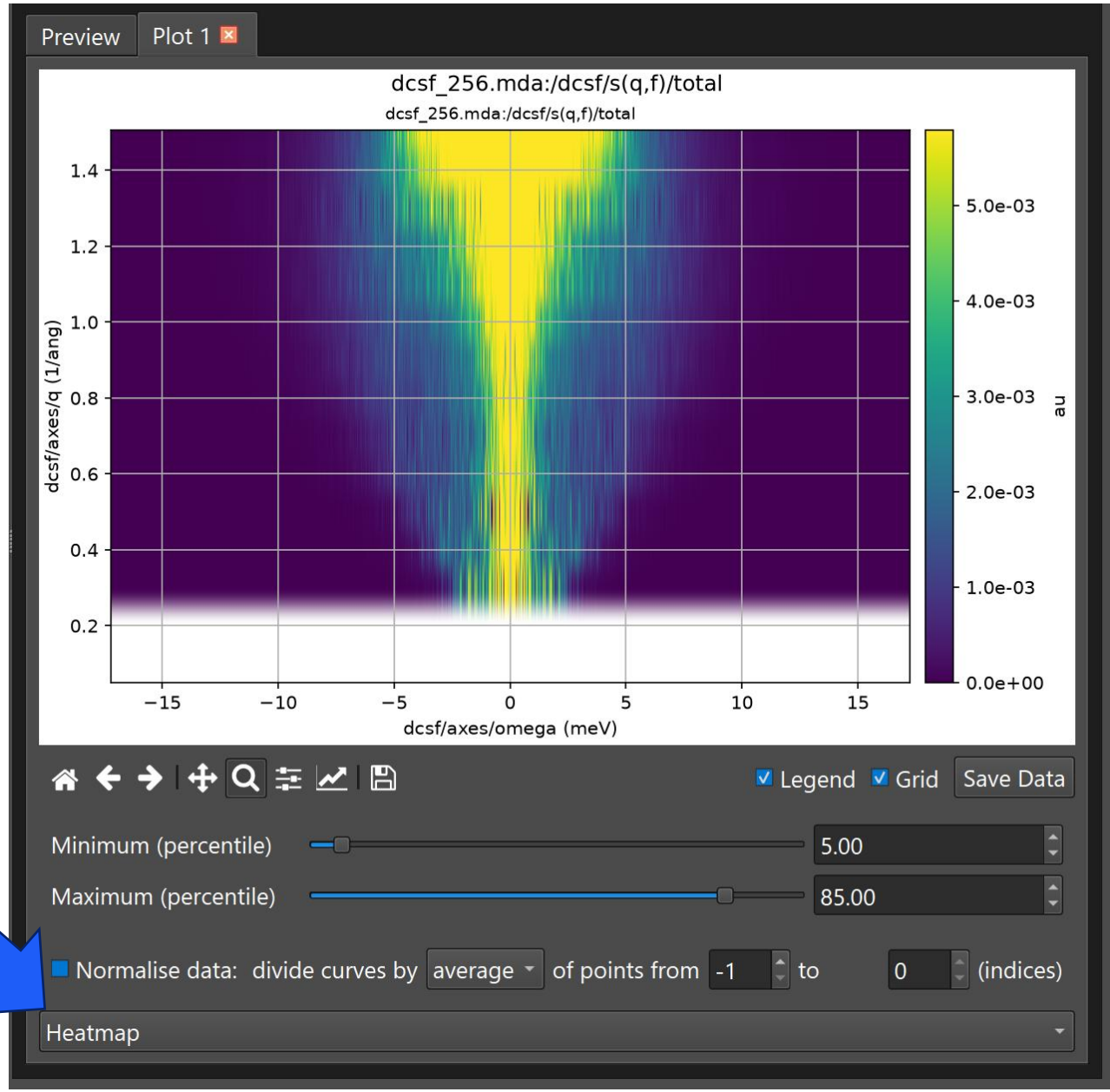
Hydrodynamics

- If you move to the vector angle statistics tab you can see the distribution of the q-vectors in more detail.



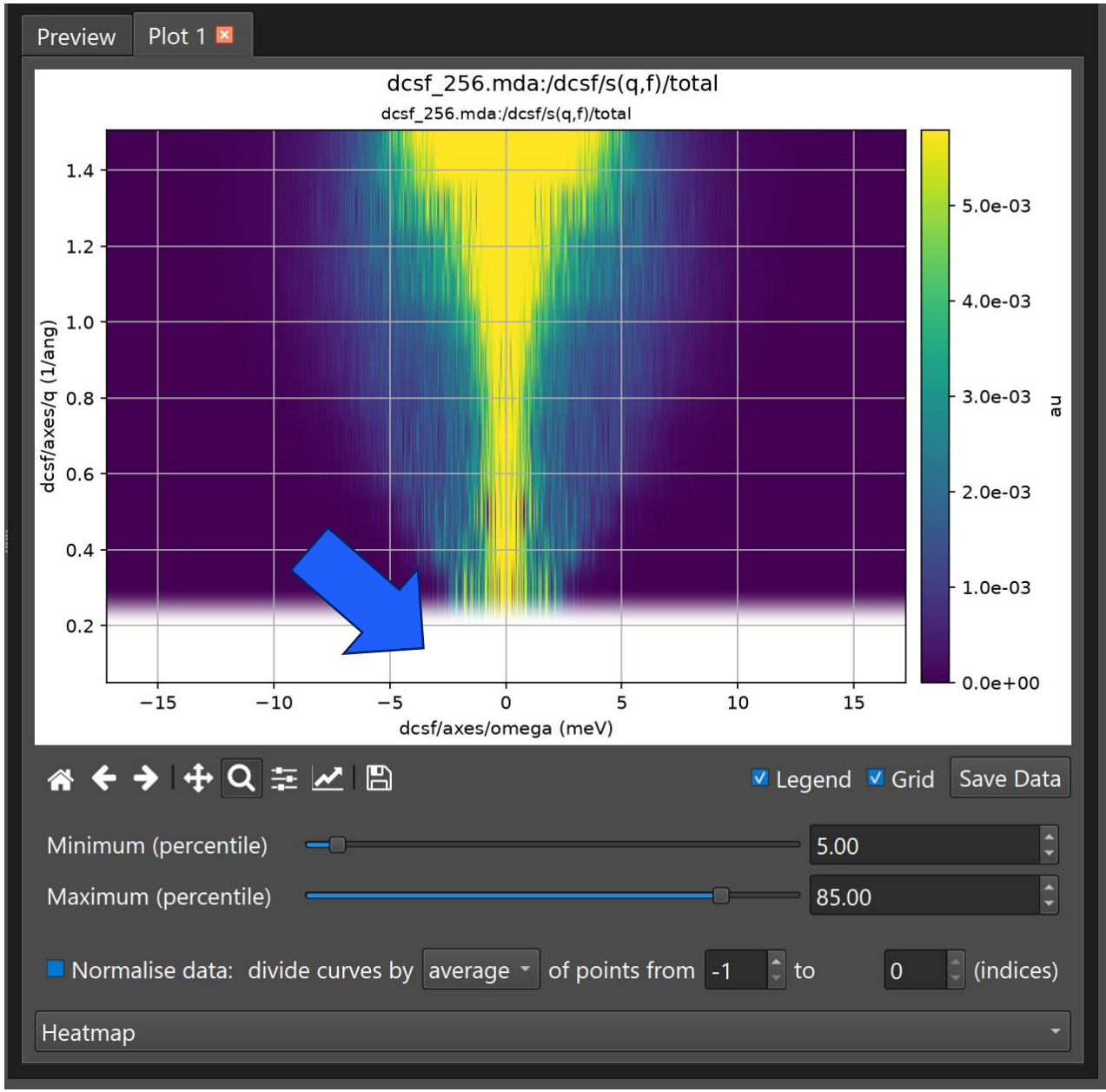
Hydrodynamics

- Let's run the calculation anyway and plots the results for `s(q,f)/total` and select `heatmap`.



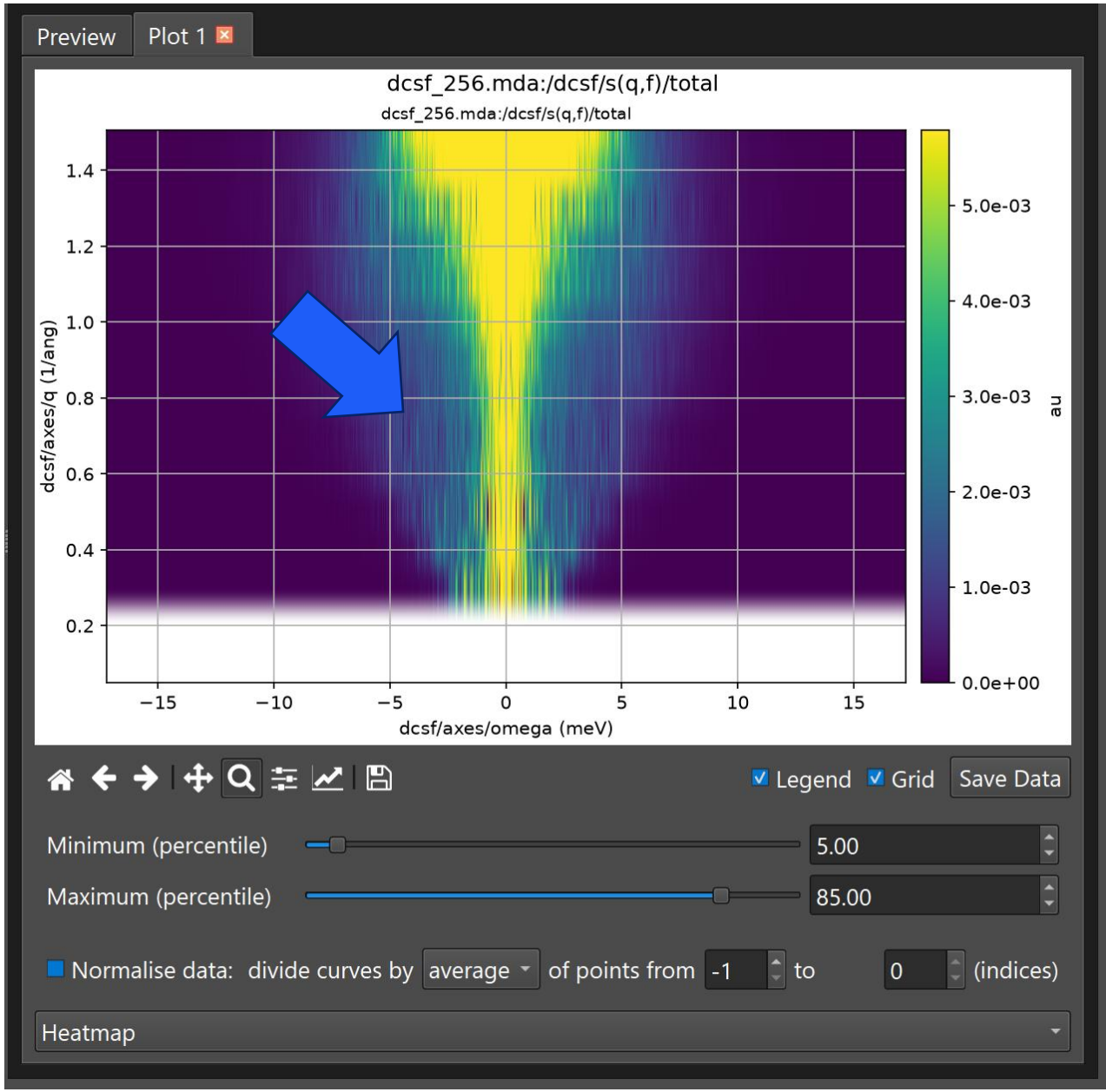
Hydrodynamics

- Let's run the calculation anyway and plots the results for `s(q,f)/total` and select `heatmap`.
- Notice that a band of results are missing, these are the empty shells.



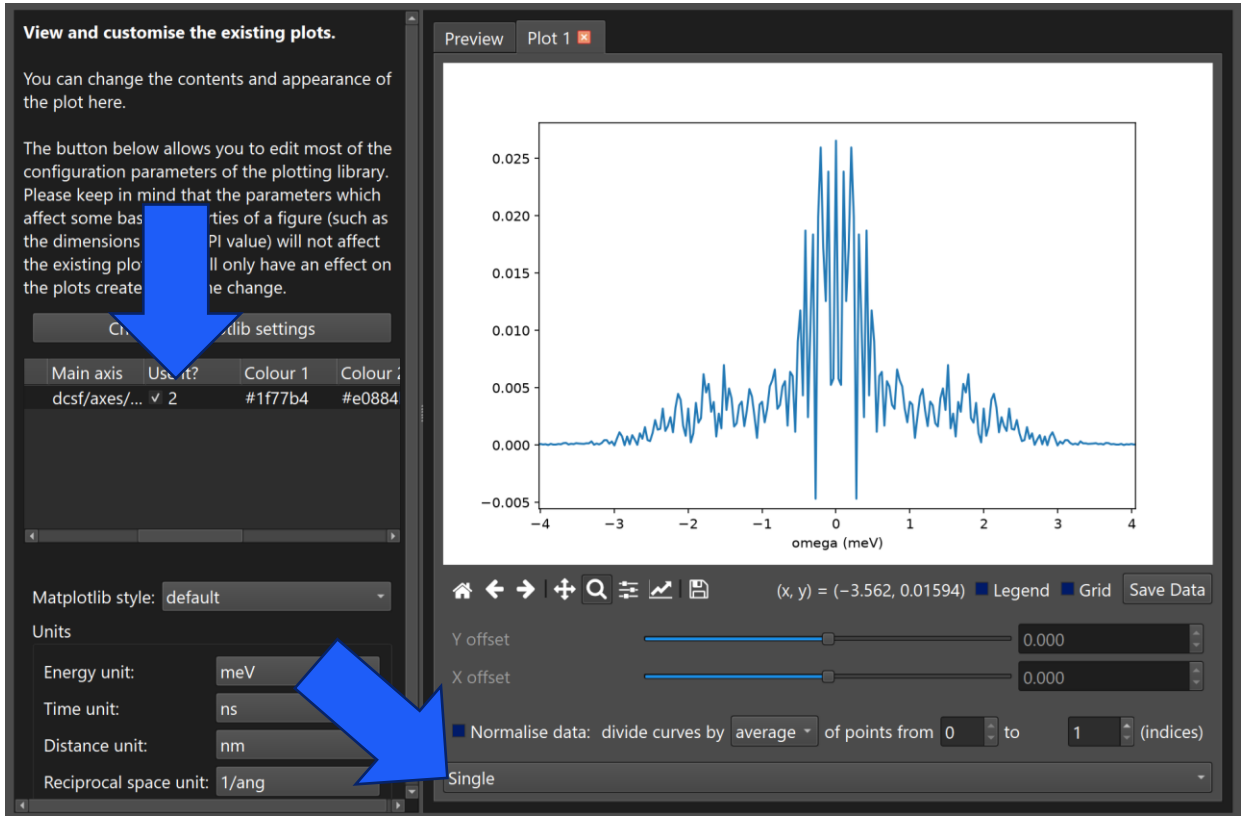
Hydrodynamics

- Let's run the calculation anyway and plots the results for `s(q,f)/total` and select `heatmap`.
- Notice that a band of results are missing, these are the empty shells.
- As our simulation was quite short and our system was quite short the results will be quite noisy.



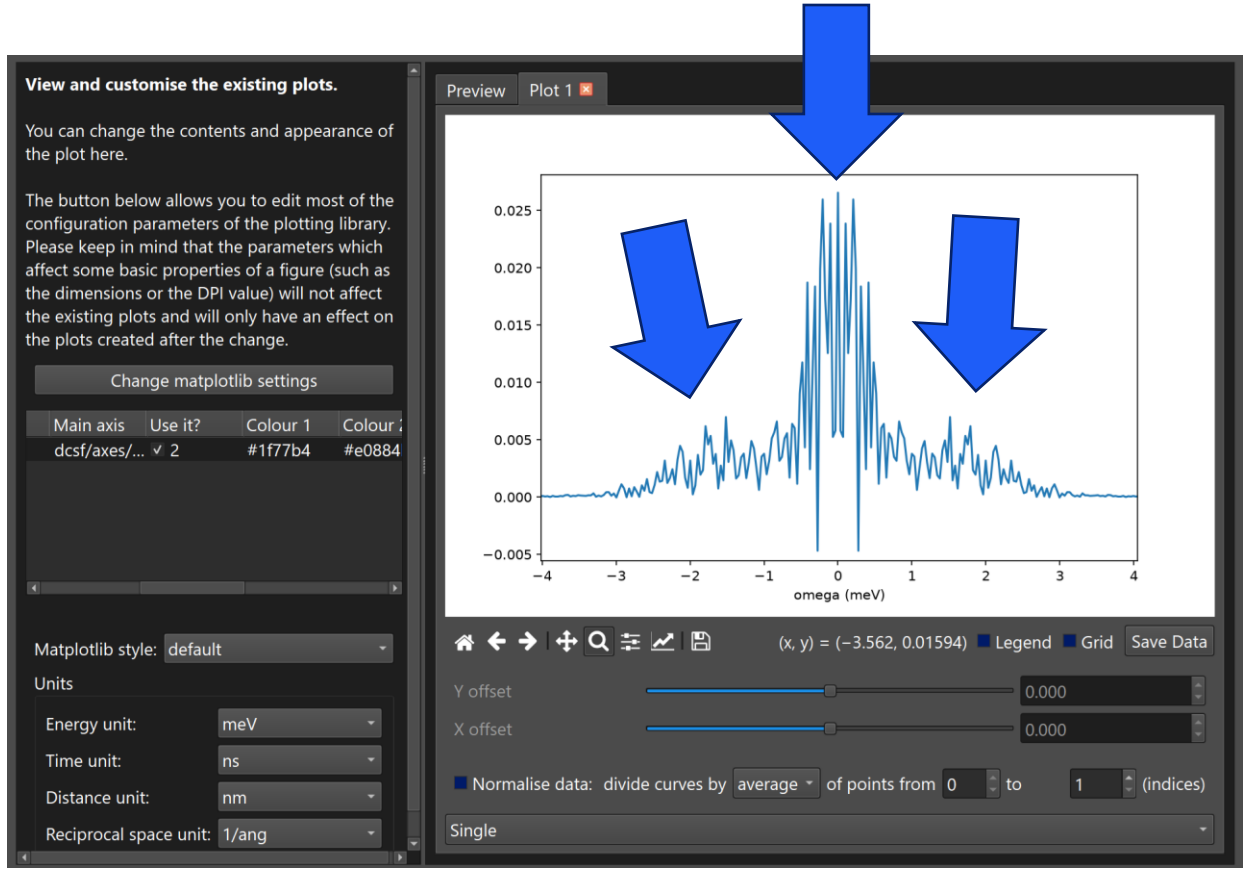
Hydrodynamics

- Select the `single` plot and plot the smallest shell we have results for this should be 2.



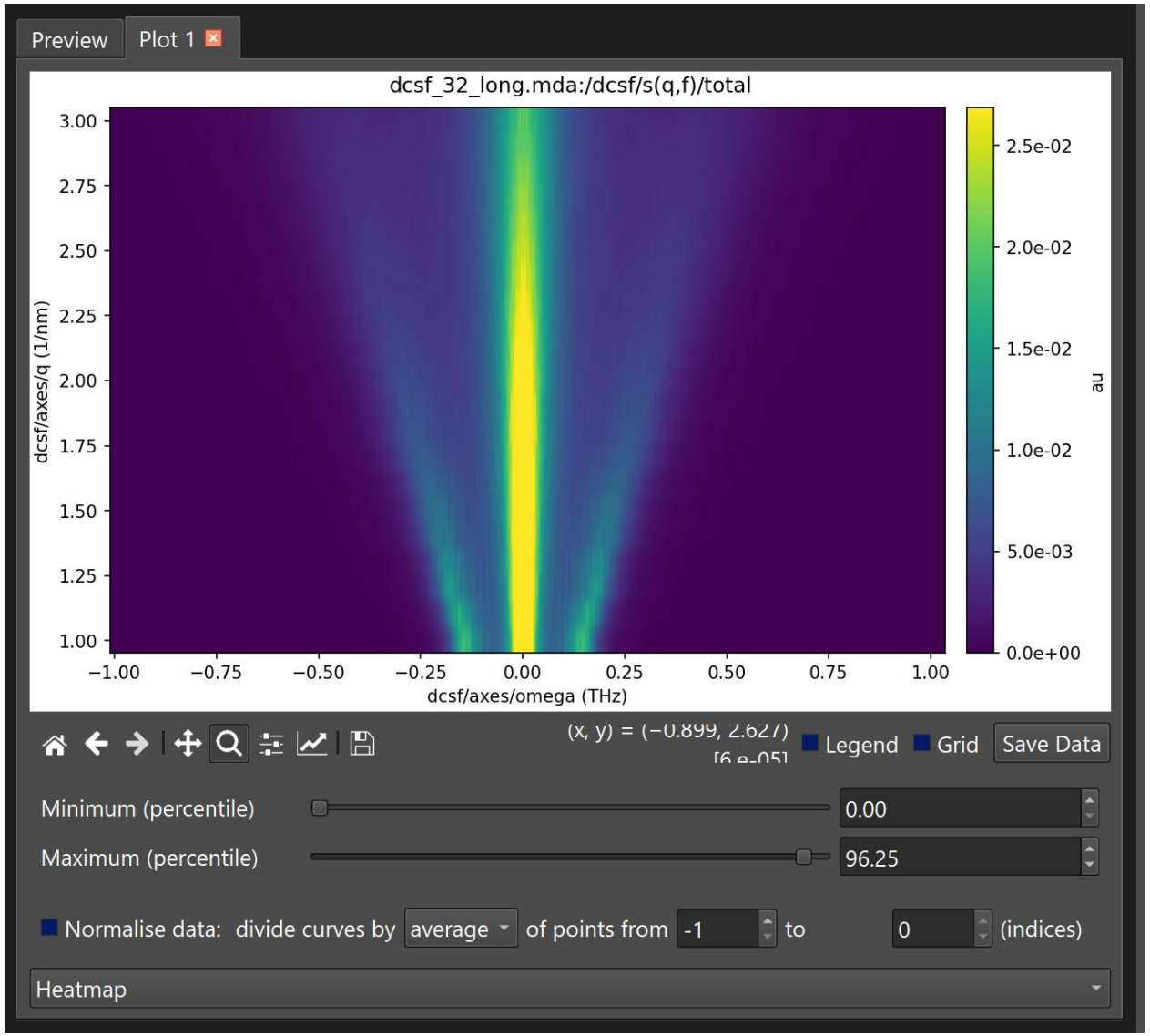
Hydrodynamics

- Select the `single` plot and plot the smallest shell we have results for this should be 2.
- Results are very noisy, but the results do seem to suggest a central Rayleigh peak and the Brillouin doublet.
- However, we need a longer trajectory and larger system to get better results.



Hydrodynamics

- In the files load up the MDA file `dcsf_32_long.mda`, this is a similar DCSF calculation but a different q-vector range and width and a larger system (131072 atoms) and a longer simulation (20000 frames with 1001 correlation frames).



Hydrodynamics

- Using the smallest value of $S(q,\omega)$ try to estimate the speed of sound. The experimental sound speed of liquid argon at a temperature of 85 K and a pressure of 592.5 mm Hg was determined to be around 850 m/s (Fleury and Boon, 1969) with light scattering.
- How close was your result? However, note that in the derivation of $S(q,\omega)$ from linear hydrodynamics the following assumptions $\gamma D_T q^2 \ll c_s q$ and $\Gamma q^2 \ll c_s q$ were made.

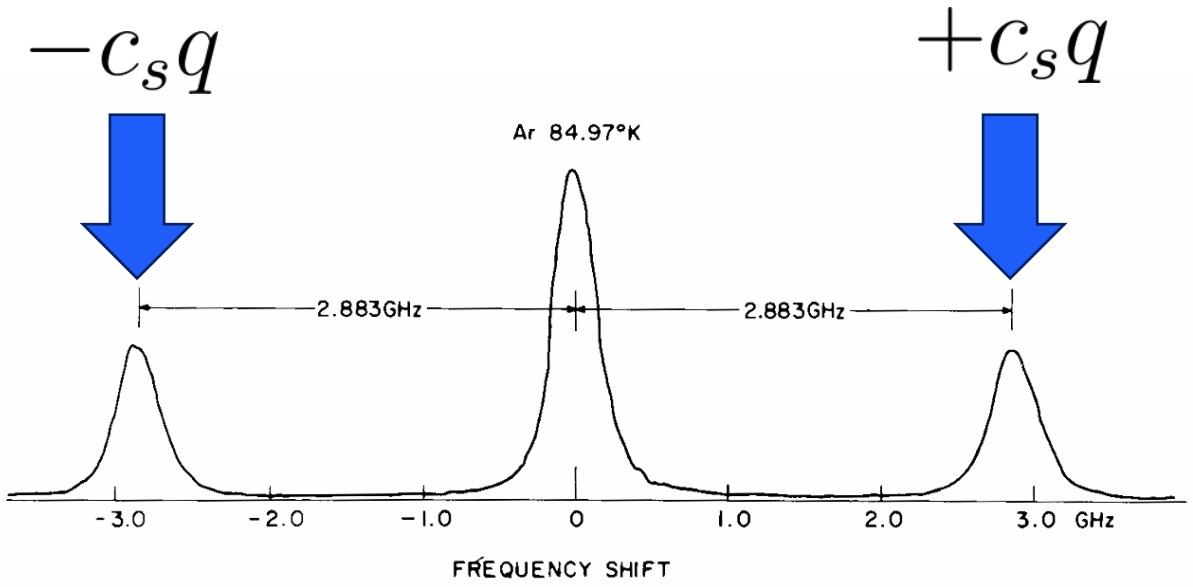
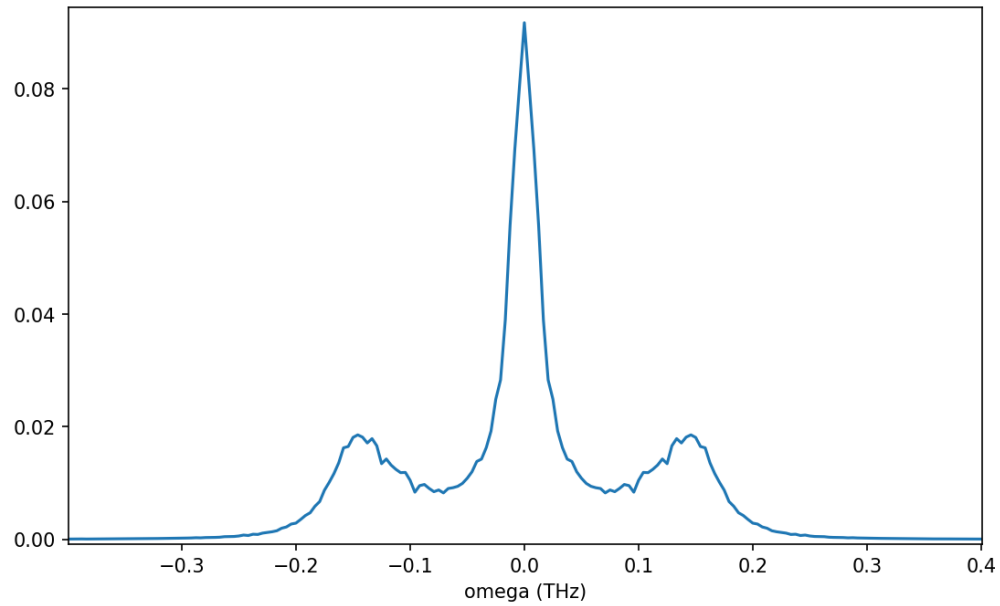


FIG. 10.4.1. Brillouin spectrum of liquid argon ($T = 84.97°\text{K}$, $\theta = 90° 14'$, laser wavelength = 5145 Å). (From Fleury and Boon, 1969.)



Neutron Brillouin Scattering

PHYSICAL REVIEW A VOLUME 11, NUMBER 1 JANUARY 1975

Neutron Brillouin scattering in fluid neon

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(Received 8 July 1974)

The results of neutron scattering experiments with scattering vectors $0.06 \leq Q \leq 0.14$ Å^{-1} and $0.27 \leq Q \leq 1.50$ Å^{-1} are discussed in terms of hydrodynamics. It is found that the small-Q data, at low densities, exhibit a distinct Brillouin triplet, describable by hydrodynamic theory. At larger Q, and at higher density, a hydrodynamic description is still possible with simple Q and ω dependences of the transport coefficents, namely a heat diffusion $D_T = D_T^0(1+Q^2 r_0^2)^{-1}$ and a viscosity $\phi = \phi^0 \tau_0^{-1}[-i\omega + \tau^{-1}(Q)]^{-1}$ with $\tau(Q) = \tau_0(1+Q^2R_0^2)^{-1}$. The relaxation lengths and times for the establishment of local equilibrium are found to be on the order of a few Angstroms and 10^{-13} sec, respectively.

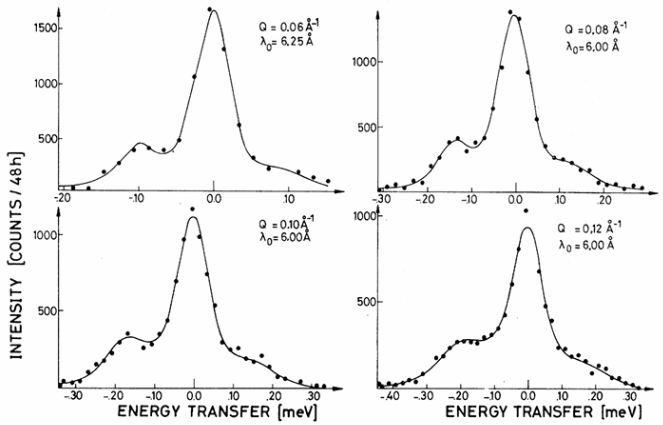


FIG. 5. Examples for fits of the hydrodynamic scattering law (7) with (8), folded with the instrumental resolution, to data obtained in the region $0.06 \leq Q \leq 0.14$ Å^{-1}; $\rho = 0.48$ g cm^{-3}, $T = 70$ K. The asymmetry of the curves results from the factor $k^3 \cot\theta_A$ and from focusing effects in the resolution function.

Neutron Brillouin scattering and *ab initio* simulation study of the collective dynamics of liquid silver

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We present a thorough investigation of the collective dynamics of liquid Ag combining neutron Brillouin scattering and *ab initio* molecular dynamics (AIMD) determinations of the dynamic structure factor $S(Q,\omega)$. The main scope of this work is not only to provide experimental results for some important dynamical properties of this liquid metal in the wave-vector range $4 < Q < 16$ nm^{-1}, but also to inquire about the scarce detectability of shear waves apparently characterizing two elements of group IB, differently from other metals. In fact, as in the case of Au, a transverse-like dynamics is not deducible from the experimental $S(Q,\omega)$ of Ag, despite the indisputable quality of the neutron data collected on the BRISP spectrometer at the Institut Laue Langevin in Grenoble. However, the significant agreement between experiment and AIMD calculations allowed for an in-depth study of the simulated $S(Q,\omega)$ in a Q range overlapping and extending the experimental one. A multimode analysis, already proven very successful in the description of various dynamical properties of fluid systems, is shown to be extremely effective also to analyze the intermediate scattering function predicted by AIMD at the various Q values, and eventually enables a reliable determination of both longitudinal and transverse branches in the dispersion curve of this liquid. Throughout the paper we highlight the importance of referring to theoretically well-founded models for $S(Q,\omega)$ and of imposing physical constraints in a fit-based analysis: These ensure that the used models obey fundamental properties of the dynamic structure factor.

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